

Digital Media Programming

Computing and Mathematics

May, 2026



Digital Media Programming (A13951)

Short Title:	Digital Media Programming	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Media Production	
Author	CDUNPHY	
Credits	5	Level: Intermediate

Description of Module / Aims

This module builds on strengths already gained in programming, digital media, and user experience. Students will build platform appropriate/independent UIs to develop software which uses or interacts with digital media by using relevant tools, software libraries, software frameworks, and environments.

Programmes

COMP -0667 BSc (Hons) in Applied Computing (WD_KCOMP_B)

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Indicative Content

- Interaction design patterns: visual hierarchy; navigational distance
- Choosing interaction styles and interaction techniques
- Presenting information: navigation; representation; manipulation
- Modern UI libraries and frameworks
- Social and digital media APIs & frameworks
- Programmatic drawing; geometry; animation
- Programmatic image processing
- Programmatic sound & video processing
- Cross platform development
- Optimization for resource-constrained devices

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Differentiate User Interfaces across multiple platforms. (Remembering).
2. Design user interface consistent with digital media programming concepts (Understanding).
3. Build a simple application with a modern graphical user interface (Creating).
4. Build an interactive application utilising digital media libraries and frameworks. (Creating).

Learning and Teaching Methods

- This module will be characterised by student participation in both formal theoretical and practical classroom activities in which students learn theoretical concepts and apply them immediately using dedicated hardware and software.
- Studio classes will be delivered in two 2-hour blocks.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	20%	1,2
<i>Project</i>	30%	3
<i>Project</i>	50%	4

Assessment Criteria

<40%: Unable to interpret and describe key concepts of digital media programming.

40%-49%: Be able to interpret and describe key concepts of digital media programming.

50%-59%: Ability to discuss key concepts of media programming and ability to discover and integrate related knowledge in other knowledge domains.

60%-69%: Be able to solve problems within digital media programming by experimenting with the appropriate skills and tools.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	48	
Independent Learning	87	

Supplementary Material

- "Adobe TV." <http://tv.adobe.com>
- "lynda.com." www.lynda.com

Requested Resources

- Room Type: Computer Lab
- COMPUTER LAB: Multimedia Lab
- EQUIPMENT: MAC PCs